KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2020
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 1.5 hours FULL MARKS: 120

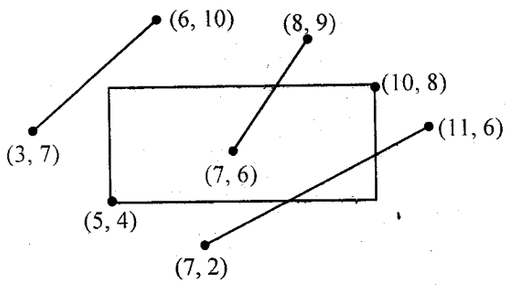
N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Find mirror reflection of a point $P(3, 3, 3)$ about a plane defined by the vertices (1, 0, 0), (0, 1, 0) and (0, 0, 1). (14)
   b) Let a View-Coordinate System (VCS) point $P(1, 1, 7)$. Find the Normalized Device Coordinate System (NDCS) value of P, if the view volume is defined by the following parameters and projection reference point is at VCS origin: (16)
   $left = -1$, $right = 1.2$, $top = 1$, $bottom = -0.8$, $near = 1$, $far = 8$.

2. a) Using Liang-barsky method clip the lines against the clipping rectangle shown in figure below. (16)



   b) Find the parallel projection of a point $P(5, 7, 8)$ with DOP, $\bar{v} = I + 2J + 2K$ onto a plane with plane normal $N = K$, and $P_0(0, 0, 1)$. (14)

3. a) Let a plane defined by the points (0, 0, 0), (0, 0, 2) and (5, 1, 1) is illuminated with a directional white light with direction vector (3, –3, 1). Find the diffuse color for all vertices and specular color for the last vertex by using Phong illumination model. Consider following parameters: (15)
   diffuse reflectivity = (0.4, 0.6, 0.1), *specular reflectivity* = (0.8, 0.9, 0.9), *shininess* = 8, *viewer position* = (6, 2, 3), *vertex normal* = *surface normal*, *no ambient light* and *no attenuation*.
   b) Determine if a ray with $s = I + J + 2K$ and $d = -2K$ intersects the triangle defined by the points (1, 0, 1), (0, 2, 1), (0, 1, 2). Use Badouel's algorithm to verify. (15)

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Define horizontal and vertical retrace. (06)
   b) How does a CRT monitor display color pictures? Briefly explain. (09)
   c) If a circle's origin is centered at point (10, 10) and radius is equivalent to 8, then rasterize the circle using Bresenham's circle drawing algorithm. (15)

5. a) Equation of two bezier curves Q and R are given as follows: (08)

$$Q(t) = (1 + t^2)P_1 + (5t^2 - 5t)P_2 + t^2P_3$$

$$R(t) = t^2P_1 + (5t^2 - 5t)P_2 + (1 + t^2)P_3$$

For both curves $0 \le t \le 1$.

Are both curves C^0 continuous at $Q(0)$ and $R(1)$. Are they both C^1 and G^1 continuous at $Q(0)$ and $R(0)$? Why or why not? Explain your answer.

   b) Find the equation of a Hermite cubic spline with the two end points as (1, 1) & (6, 5) and corresponding tangent vectors as (0, 4) & (4, 0). (13)
   c) How is Koch curves generated? Explain with suitable example(s)? (09)

6. a) How is painting priority of a surface determined by painter's algorithm? (08)
   b) Explain z-buffer algorithm in detail for visible surface detection. What are the advantages and disadvantages of z-buffer algorithm? (13)
   c) Define texture minification. Why is mipmapping used in computer graphics? Briefly explain. (09)